$$
\begin{aligned}
& C_\phi^{(22)\ (21)\ (12)} \to \frac{1}{16\pi^2}\Big(-\frac{1}{9} g_2^6\, \mathrm{LF}_{3,0}[m_2] - \frac{1}{6} g_2^6\, \mathrm{LF}_{4,-1}[m_2] + \frac{8}{45} g_2^6\, \mathrm{LF}_{5,-2}[m_2] + \\
& \frac{1}{72}\Big(18 g_2^2 \overline{y}_e^{pr} y_e^{pr}\left(g_1^2-3g_2^2\right) + \textstyle\sum_p g_2^4\left(-9g_1^2+5g_2^2\right)\Big)\, \mathrm{LF}_{3,0}[m_{\tilde l}^{p}] + \\
& \Big(\frac12 g_2^4 \overline{y}_e^{pr} y_e^{pr} - \frac{1}{48}\textstyle\sum_p g_2^6\Big)\, \mathrm{LF}_{4,-1}[m_{\tilde l}^{p}] - \frac{2}{45}\textstyle\sum_p g_2^6\, \mathrm{LF}_{5,-2}[m_{\tilde l}^{p}] + \\
& \frac{1}{24} g_2^2\Big(-6\overline{y}_d^{pr} y_d^{pr}\left(g_1^2+9g_2^2\right) - 6g_1^2\overline{y}_u^{pr} y_u^{pr} + \textstyle\sum_p g_2^2\left(3g_1^2+5g_2^2\right)\Big)\, \mathrm{LF}_{3,0}[m_{\tilde q}^{p}] + \\
& \Big(\frac32 g_2^4 \overline{y}_d^{pr} y_d^{pr} - \frac{1}{16}\textstyle\sum_p g_2^6\Big)\, \mathrm{LF}_{4,-1}[m_{\tilde q}^{p}] - \frac{2}{15}\textstyle\sum_p g_2^6\, \mathrm{LF}_{5,-2}[m_{\tilde q}^{p}] - \\
& \frac{1}{18} g_2^6\, \mathrm{LF}_{3,0}[\tilde\mu] - \frac{1}{12} g_2^6\, \mathrm{LF}_{4,-1}[\tilde\mu] + \frac{4}{45} g_2^6\, \mathrm{LF}_{5,-2}[\tilde\mu] - g_1^6\tilde\mu^2\, \mathrm{LF}_{3,3,-2}[m_1,\tilde\mu] - \\
& g_1^6 m_1^2\tilde\mu^2\, \mathrm{LF}_{3,3,-1}[m_1,\tilde\mu] + \frac23 g_2^6\, \mathrm{LF}_{2,1,0}[m_2,\tilde\mu] + \frac{17}{12} g_2^6\, \mathrm{LF}_{2,2,-1}[m_2,\tilde\mu] - \\
& \frac13 g_2^6\, \mathrm{LF}_{3,1,-1}[m_2,\tilde\mu] - \frac73 g_2^6\, \mathrm{LF}_{3,2,-2}[m_2,\tilde\mu] + \frac32 g_2^6 m_2^2\, \mathrm{LF}_{3,2,-1}[m_2,\tilde\mu] + \\
& 3g_2^6\, \mathrm{LF}_{3,3,-3}[m_2,\tilde\mu] - 3g_2^6\left(m_2^2+\tilde\mu^2\right) \mathrm{LF}_{3,3,-2}[m_2,\tilde\mu] - 3g_2^6 m_2^2\tilde\mu^2\, \mathrm{LF}_{3,3,-1}[m_2,\tilde\mu] + \\
& g_2^6\, \mathrm{LF}_{4,2,-3}[m_2,\tilde\mu] - g_2^6 m_2^2\, \mathrm{LF}_{4,2,-2}[m_2,\tilde\mu] - \frac12 g_1^2 g_2^2\overline{a}_d^{pr} a_d^{pr}\, \mathrm{LF}_{2,2,0}[m_{\tilde d}^{r}, m_{\tilde q}^{p}] + \\
& \frac12 g_1^2\overline{y}_d^{pr} y_d^{sr}\overline{y}_u^{st} y_u^{pt}\, \mathrm{LF}_{2,1,0}[m_{\tilde d}^{r}, m_{\tilde u}^{t}] - \frac12 g_1^2 g_2^2\overline{a}_e^{pr} a_e^{pr}\, \mathrm{LF}_{2,2,0}[m_{\tilde e}^{r}, m_{\tilde l}^{p}] - \\
& g_2^4\overline{a}_e^{pr} a_e^{pr}\, \mathrm{LF}_{3,1,0}[m_{\tilde l}^{p}, m_{\tilde e}^{r}] + \frac14 g_1^2 g_2^2\left(\overline{a}_e^{pr} a_e^{pr} + \tilde\mu^2\overline{y}_e^{pr} y_e^{pr}\right) \mathrm{LF}_{3,2,-1}[m_{\tilde l}^{p}, m_{\tilde e}^{r}] + \\
& \frac14 g_2^2\left(\overline{a}_e^{pr} a_e^{pr}\left(3g_1^2+4g_2^2\right) - 3g_1^2\tilde\mu^2\overline{y}_e^{pr} y_e^{pr}\right) \mathrm{LF}_{4,1,-1}[m_{\tilde l}^{p}, m_{\tilde e}^{r}] - \\
& \frac16 g_2^2\left(\overline{a}_e^{pr} a_e^{pr}\left(3g_1^2+g_2^2\right) - 3g_1^2\tilde\mu^2\overline{y}_e^{pr} y_e^{pr}\right) \mathrm{LF}_{5,1,-2}[m_{\tilde l}^{p}, m_{\tilde e}^{r}] + \\
& g_2^2\overline{y}_e^{pr}\overline{y}_e^{st} y_e^{pt} y_e^{sr}\, \mathrm{LF}_{2,1,0}[m_{\tilde l}^{p}, m_{\tilde l}^{s}] - \frac12 g_2^2\overline{y}_e^{pr}\overline{y}_e^{st} y_e^{pt} y_e^{sr}\, \mathrm{LF}_{3,1,-1}[m_{\tilde l}^{p}, m_{\tilde l}^{s}] - \\
& g_2^2\overline{a}_d^{pr} a_d^{pr}\left(2g_1^2+3g_2^2\right) \mathrm{LF}_{3,1,0}[m_{\tilde q}^{p}, m_{\tilde d}^{r}] + \\
& \frac14 g_1^2 g_2^2\left(\overline{a}_d^{pr} a_d^{pr} + \tilde\mu^2\overline{y}_d^{pr} y_d^{pr}\right) \mathrm{LF}_{3,2,-1}[m_{\tilde q}^{p}, m_{\tilde d}^{r}] + \\
& \frac34 g_2^2\left(\overline{a}_d^{pr} a_d^{pr}\left(5g_1^2+4g_2^2\right) - g_1^2\tilde\mu^2\overline{y}_d^{pr} y_d^{pr}\right) \mathrm{LF}_{4,1,-1}[m_{\tilde q}^{p}, m_{\tilde d}^{r}] - \\
& \frac12 g_2^2\left(\overline{a}_d^{pr} a_d^{pr}\left(3g_1^2+g_2^2\right) - 3g_1^2\tilde\mu^2\overline{y}_d^{pr} y_d^{pr}\right) \mathrm{LF}_{5,1,-2}[m_{\tilde q}^{p}, m_{\tilde d}^{r}] + \\
& \frac14\overline{y}_d^{pr} y_d^{sr}\left(12g_2^2\overline{y}_d^{st} y_d^{pt} + \overline{y}_u^{st} y_u^{pt}\left(g_1^2+3g_2^2\right)\right) \mathrm{LF}_{2,1,0}[m_{\tilde q}^{p}, m_{\tilde q}^{s}] - \\
& \frac32 g_2^2\overline{y}_d^{pr}\overline{y}_d^{st} y_d^{pt} y_d^{sr}\, \mathrm{LF}_{3,1,-1}[m_{\tilde q}^{p}, m_{\tilde q}^{s}] + \\
& \frac14 g_2^2\left(3g_1^2\overline{a}_u^{pr} a_u^{pr} + \tilde\mu^2\overline{y}_u^{pr} y_u^{pr}\left(-g_1^2+g_2^2\right)\right) \mathrm{LF}_{2,2,0}[m_{\tilde q}^{p}, m_{\tilde u}^{r}] + \\
& \frac14 g_2^2\left(-g_1^2\overline{a}_u^{pr} a_u^{pr} + \tilde\mu^2\overline{y}_u^{pr} y_u^{pr}\left(g_1^2+3g_2^2\right)\right) \mathrm{LF}_{3,1,0}[m_{\tilde q}^{p}, m_{\tilde u}^{r}] - \\
& \frac12 g_2^4\tilde\mu^2\overline{y}_u^{pr} y_u^{pr}\, \mathrm{LF}_{3,2,-1}[m_{\tilde q}^{p}, m_{\tilde u}^{r}] - \frac32 g_2^4\tilde\mu^2\overline{y}_u^{pr} y_u^{pr}\, \mathrm{LF}_{4,1,-1}[m_{\tilde q}^{p}, m_{\tilde u}^{r}] + \\
& \frac14\overline{y}_d^{pr} y_d^{sr}\overline{y}_u^{st} y_u^{pt}\left(g_1^2+3g_2^2\right) \mathrm{LF}_{2,1,0}[m_{\tilde q}^{s}, m_{\tilde q}^{p}] + \\
& \frac14 g_2^2\left(7g_1^2\overline{a}_u^{pr} a_u^{pr} + \tilde\mu^2\overline{y}_u^{pr} y_u^{pr}\left(g_1^2-g_2^2\right)\right) \mathrm{LF}_{3,1,0}[m_{\tilde u}^{r}, m_{\tilde q}^{p}] - \\
& \frac14 g_2^2\left(g_1^2\overline{a}_u^{pr} a_u^{pr} + \tilde\mu^2\overline{y}_u^{pr} y_u^{pr}\left(g_1^2+g_2^2\right)\right) \mathrm{LF}_{3,2,-1}[m_{\tilde u}^{r}, m_{\tilde q}^{p}] + \\
& \frac34 g_2^2\left(-5g_1^2\overline{a}_u^{pr} a_u^{pr} + \tilde\mu^2\overline{y}_u^{pr} y_u^{pr}\left(g_1^2+g_2^2\right)\right) \mathrm{LF}_{4,1,-1}[m_{\tilde u}^{r}, m_{\tilde q}^{p}] - \\
& \frac12 g_2^2\left(-3g_1^2\overline{a}_u^{pr} a_u^{pr} + \tilde\mu^2\overline{y}_u^{pr} y_u^{pr}\left(3g_1^2+g_2^2\right)\right) \mathrm{LF}_{5,1,-2}[m_{\tilde u}^{r}, m_{\tilde q}^{p}] + \\
& \frac12\overline{y}_d^{pr} y_d^{sr}\overline{y}_u^{st} y_u^{pt}\left(-2g_1^2+3g_2^2\right) \mathrm{LF}_{2,1,0}[m_{\tilde u}^{t}, m_{\tilde d}^{r}] - \\
& \frac32 g_2^2\overline{y}_d^{pr} y_d^{sr}\overline{y}_u^{st} y_u^{pt}\, \mathrm{LF}_{3,1,-1}[m_{\tilde u}^{t}, m_{\tilde d}^{r}] + \frac12 g_1^2 g_2^4\, \mathrm{LF}_{3,1,-1}[\tilde\mu, m_1] - \\
& \frac34 g_1^4 g_2^2\, \mathrm{LF}_{3,2,-2}[\tilde\mu, m_1] - \frac34 g_1^4 g_2^2 m_1^2\, \mathrm{LF}_{3,2,-1}[\tilde\mu, m_1] - \frac{7}{12} g_1^2 g_2^4\, \mathrm{LF}_{4,1,-2}[\tilde\mu, m_1] + \\
& \frac12 g_1^4 g_2^2\, \mathrm{LF}_{4,2,-3}[\tilde\mu, m_1] + \frac12 g_1^4 g_2^2 m_1^2\, \mathrm{LF}_{4,2,-2}[\tilde\mu, m_1] + \frac16 g_1^2 g_2^4\, \mathrm{LF}_{5,1,-3}[\tilde\mu, m_1] + \\
& \frac16 g_2^6\, \mathrm{LF}_{3,1,-1}[\tilde\mu, m_2] - \frac{65}{12} g_2^6\, \mathrm{LF}_{3,2,-2}[\tilde\mu, m_2] - \frac34 g_2^6 m_2^2\, \mathrm{LF}_{3,2,-1}[\tilde\mu, m_2] - \frac34 g_2^6\, \mathrm{LF}_{4,1,-2}[\tilde\mu, m_2] + \\
& 3g_2^6\, \mathrm{LF}_{4,2,-3}[\tilde\mu, m_2] + \frac12 g_2^6\left(m_2^2+\tilde\mu^2\right) \mathrm{LF}_{4,2,-2}[\tilde\mu, m_2] + \frac12 g_2^6\, \mathrm{LF}_{5,1,-3}[\tilde\mu, m_2] - \\
& 2g_1^2 g_2^4\, \mathrm{LF}_{2,2,1,-2}[m_2,\tilde\mu,m_1] - \frac32 g_1^2 g_2^4\left(m_1 m_2 + 2\tilde\mu^2\right) \mathrm{LF}_{2,2,1,-1}[m_2,\tilde\mu,m_1] + \\
& g_1^2 g_2^4\, \mathrm{LF}_{3,2,1,-3}[m_2,\tilde\mu,m_1] + g_1^2 g_2^4\left(m_1 m_2 + 2\tilde\mu^2\right) \mathrm{LF}_{3,2,1,-2}[m_2,\tilde\mu,m_1] - \\
& 3\overline{y}_d^{pr}\overline{y}_d^{st} y_d^{pt} y_d^{ur}\overline{y}_u^{uv} y_u^{sv}\, \mathrm{LF}_{1,1,1,0}[m_{\tilde d}^{r}, m_{\tilde d}^{t}, m_{\tilde u}^{v}] + \\
& \frac12 g_1^2\left(\overline{y}_u^{st} y_u^{pt}\overline{a}_d^{pr} a_d^{sr} - \tilde\mu^2\overline{y}_d^{pr}\overline{y}_d^{st} y_d^{pt} y_d^{sr}\right) \mathrm{LF}_{2,1,1,0}[m_{\tilde d}^{r}, m_{\tilde q}^{p}, m_{\tilde q}^{s}] + \\
& \frac12 g_1^2 y_d^{sr}\overline{y}_u^{st}\left(\overline{a}_d^{pr} a_u^{pt} - \tilde\mu^2\overline{y}_d^{pr} y_u^{pt}\right) \mathrm{LF}_{2,1,1,0}[m_{\tilde d}^{r}, m_{\tilde q}^{p}, m_{\tilde u}^{t}] - \\
& \frac12 g_1^2\tilde\mu^2\overline{y}_d^{pr} y_d^{sr}\overline{y}_u^{st} y_u^{pt}\, \mathrm{LF}_{2,1,1,0}[m_{\tilde d}^{r}, m_{\tilde q}^{s}, m_{\tilde u}^{t}] + \\
& \frac12 g_1^2\overline{y}_d^{st} y_u^{sr} a_d^{pt}\overline{a}_u^{pr}\, \mathrm{LF}_{2,1,1,0}[m_{\tilde d}^{t}, m_{\tilde q}^{p}, m_{\tilde u}^{r}] - \\
& \frac12 g_1^2\tilde\mu^2\overline{y}_e^{pr}\overline{y}_e^{st} y_e^{pt} y_e^{sr}\, \mathrm{LF}_{2,1,1,0}[m_{\tilde e}^{r}, m_{\tilde l}^{p}, m_{\tilde l}^{s}] + g_2^2\overline{y}_e^{st} y_e^{sr}\overline{a}_e^{pr} a_e^{pt} \\
& \quad \mathrm{LF}_{2,1,1,0}[m_{\tilde l}^{p}, m_{\tilde e}^{r}, m_{\tilde e}^{t}] - \frac12 g_2^2\overline{y}_e^{st} y_e^{sr}\overline{a}_e^{pr} a_e^{pt}\, \mathrm{LF}_{3,1,1,-1}[m_{\tilde l}^{p}, m_{\tilde e}^{r}, m_{\tilde e}^{t}] + \\
& \frac12\overline{y}_e^{st} y_e^{pt}\left(2g_2^2\overline{a}_e^{pr} a_e^{sr} + g_1^2\tilde\mu^2\overline{y}_e^{pr} y_e^{sr}\right) \mathrm{LF}_{2,1,1,0}[m_{\tilde l}^{p}, m_{\tilde e}^{r}, m_{\tilde l}^{s}] - \\
& \frac14\overline{y}_e^{st} y_e^{pt}\left(2g_2^2\overline{a}_e^{pr} a_e^{sr} + \tilde\mu^2\overline{y}_e^{pr} y_e^{sr}\left(g_1^2+g_2^2\right)\right) \mathrm{LF}_{3,1,1,-1}[m_{\tilde l}^{p}, m_{\tilde e}^{r}, m_{\tilde l}^{s}] + \\
& \frac12 g_1^2\tilde\mu^2\overline{y}_e^{pr}\overline{y}_e^{st} y_e^{pt} y_e^{sr}\, \mathrm{LF}_{2,1,1,0}[m_{\tilde l}^{p}, m_{\tilde e}^{t}, m_{\tilde l}^{s}] - \\
& \frac14\tilde\mu^2\overline{y}_e^{pr}\overline{y}_e^{st} y_e^{pt} y_e^{sr}\left(g_1^2+g_2^2\right) \mathrm{LF}_{3,1,1,-1}[m_{\tilde l}^{p}, m_{\tilde e}^{t}, m_{\tilde l}^{s}] - \\
& \frac14\tilde\mu^2\overline{y}_e^{pr}\overline{y}_e^{st} y_e^{pt} y_e^{sr}\left(g_1^2+g_2^2\right) \mathrm{LF}_{2,2,1,-1}[m_{\tilde l}^{p}, m_{\tilde l}^{s}, m_{\tilde e}^{r}] - \frac12 g_2^2\tilde\mu^2\overline{y}_e^{pr}\overline{y}_e^{st} y_e^{pt} \\
& \quad y_e^{sr}\, \mathrm{LF}_{3,1,1,-1}[m_{\tilde l}^{s}, m_{\tilde e}^{r}, m_{\tilde e}^{t}] + g_2^2\overline{y}_e^{st} y_e^{pt}\overline{a}_e^{pr} a_e^{sr}\, \mathrm{LF}_{2,1,1,0}[m_{\tilde l}^{s}, m_{\tilde e}^{r}, m_{\tilde l}^{p}] - \\
& \frac12 g_2^2\overline{y}_e^{st} y_e^{pt}\overline{a}_e^{pr} a_e^{sr}\, \mathrm{LF}_{3,1,1,-1}[m_{\tilde l}^{s}, m_{\tilde e}^{r}, m_{\tilde l}^{p}] + 3g_2^2\overline{y}_d^{st} y_d^{sr}\overline{a}_d^{pr} a_d^{pt} \\
& \quad \mathrm{LF}_{2,1,1,0}[m_{\tilde q}^{p}, m_{\tilde d}^{r}, m_{\tilde d}^{t}] - \frac32 g_2^2\overline{y}_d^{st} y_d^{sr}\overline{a}_d^{pr} a_d^{pt}\, \mathrm{LF}_{3,1,1,-1}[m_{\tilde q}^{p}, m_{\tilde d}^{r}, m_{\tilde d}^{t}] + \\
& \frac12\left(\overline{a}_d^{pr} a_d^{sr}\left(6g_2^2\overline{y}_d^{st} y_d^{pt} + \overline{y}_u^{st} y_u^{pt}\left(2g_1^2+3g_2^2\right)\right) + g_1^2\tilde\mu^2\overline{y}_d^{pr}\overline{y}_d^{st} y_d^{pt} y_d^{sr}\right) \\
& \quad \mathrm{LF}_{2,1,1,0}[m_{\tilde q}^{p}, m_{\tilde d}^{r}, m_{\tilde q}^{s}] - \frac34\Big(\overline{a}_d^{pr} a_d^{sr}\left(2g_2^2\overline{y}_d^{st} y_d^{pt} + \overline{y}_u^{st} y_u^{pt}\left(g_1^2+g_2^2\right)\right) + \\
& \qquad \tilde\mu^2\overline{y}_d^{pr}\overline{y}_d^{st} y_d^{pt} y_d^{sr}\left(g_1^2+g_2^2\right)\Big) \mathrm{LF}_{3,1,1,-1}[m_{\tilde q}^{p}, m_{\tilde d}^{r}, m_{\tilde q}^{s}] + \\
& \frac12 y_d^{sr}\overline{y}_u^{st}\left(\overline{a}_d^{pr} a_u^{pt}\left(2g_1^2+3g_2^2\right) + g_1^2\tilde\mu^2\overline{y}_d^{pr} y_u^{pt}\right) \mathrm{LF}_{2,1,1,0}[m_{\tilde q}^{p}, m_{\tilde d}^{r}, m_{\tilde u}^{t}] - \\
& \frac34 y_d^{sr}\overline{y}_u^{st}\left(\overline{a}_d^{pr} a_u^{pt}\left(g_1^2+g_2^2\right) + \tilde\mu^2\overline{y}_d^{pr} y_u^{pt}\left(g_1^2-g_2^2\right)\right) \mathrm{LF}_{3,1,1,-1}[m_{\tilde q}^{p}, m_{\tilde d}^{r}, m_{\tilde u}^{t}] + \\
& \frac12 g_1^2\tilde\mu^2\overline{y}_d^{pr}\overline{y}_d^{st} y_d^{pt} y_d^{sr}\, \mathrm{LF}_{2,1,1,0}[m_{\tilde q}^{p}, m_{\tilde d}^{t}, m_{\tilde q}^{s}] - \\
& \frac34\tilde\mu^2\overline{y}_d^{pr}\overline{y}_d^{st} y_d^{pt} y_d^{sr}\left(g_1^2+g_2^2\right) \mathrm{LF}_{3,1,1,-1}[m_{\tilde q}^{p}, m_{\tilde d}^{t}, m_{\tilde q}^{s}] + \\
& \frac12\overline{y}_d^{st} y_u^{sr} a_d^{pt}\overline{a}_u^{pr}\left(2g_1^2+3g_2^2\right) \mathrm{LF}_{2,1,1,0}[m_{\tilde q}^{p}, m_{\tilde d}^{t}, m_{\tilde u}^{r}] - \\
& \frac34\overline{y}_d^{st} y_u^{sr} a_d^{pt}\overline{a}_u^{pr}\left(g_1^2+g_2^2\right) \mathrm{LF}_{3,1,1,-1}[m_{\tilde q}^{p}, m_{\tilde d}^{t}, m_{\tilde u}^{r}] - \\
& \frac34\left(g_1^2+g_2^2\right)\left(\overline{y}_u^{st} y_u^{pt}\overline{a}_d^{pr} a_d^{sr} + \tilde\mu^2\overline{y}_d^{pr}\overline{y}_d^{st} y_d^{pt} y_d^{sr}\right) \mathrm{LF}_{2,2,1,-1}[m_{\tilde q}^{p}, m_{\tilde q}^{s}, m_{\tilde d}^{r}] - \\
& 3\overline{y}_d^{pr}\overline{y}_d^{st} y_d^{pt} y_d^{ur}\overline{y}_u^{uv} y_u^{sv}\, \mathrm{LF}_{1,1,1,0}[m_{\tilde q}^{p}, m_{\tilde q}^{s}, m_{\tilde q}^{u}] + \frac14 \\
& \quad \left(\overline{y}_d^{st} y_d^{pt}\overline{a}_u^{pr} a_u^{sr}\left(g_1^2+3g_2^2\right) - \tilde\mu^2\overline{y}_u^{pr}\overline{y}_u^{st} y_u^{pt} y_u^{sr}\left(g_1^2-3g_2^2\right)\right) \mathrm{LF}_{2,1,1,0}[m_{\tilde q}^{p}, m_{\tilde q}^{s}, m_{\tilde u}^{r}] - \\
& \frac14\tilde\mu^2\overline{y}_u^{st} y_u^{pt}\left(6g_2^2\overline{y}_d^{pr} y_d^{sr} + \overline{y}_u^{pr} y_u^{sr}\left(g_1^2-3g_2^2\right)\right) \mathrm{LF}_{2,1,1,0}[m_{\tilde q}^{p}, m_{\tilde q}^{s}, m_{\tilde u}^{t}] + \\
& \frac32 g_2^2\tilde\mu^2\overline{y}_d^{pr} y_d^{sr}\overline{y}_u^{st} y_u^{pt}\, \mathrm{LF}_{3,1,1,-1}[m_{\tilde q}^{p}, m_{\tilde q}^{s}, m_{\tilde u}^{t}] - \\
& \frac34\overline{y}_d^{st} y_u^{sr} a_d^{pt}\overline{a}_u^{pr}\left(g_1^2+g_2^2\right) \mathrm{LF}_{2,2,1,-1}[m_{\tilde q}^{p}, m_{\tilde u}^{r}, m_{\tilde d}^{t}] - \\
& \frac34 y_d^{sr}\overline{y}_u^{st}\left(\overline{a}_d^{pr} a_u^{pt}\left(g_1^2+g_2^2\right) + \tilde\mu^2\overline{y}_d^{pr} y_u^{pt}\left(g_1^2-g_2^2\right)\right) \mathrm{LF}_{2,2,1,-1}[m_{\tilde q}^{p}, m_{\tilde u}^{t}, m_{\tilde d}^{r}] + \\
& \frac34 g_2^2\tilde\mu^2\overline{y}_d^{pr} y_d^{sr}\overline{y}_u^{st} y_u^{pt}\, \mathrm{LF}_{2,2,1,-1}[m_{\tilde q}^{p}, m_{\tilde u}^{t}, m_{\tilde q}^{s}] - \\
& \frac32 g_2^2\tilde\mu^2\overline{y}_d^{pr}\overline{y}_d^{st} y_d^{pt} y_d^{sr}\, \mathrm{LF}_{3,1,1,-1}[m_{\tilde q}^{s}, m_{\tilde d}^{r}, m_{\tilde d}^{t}] + \\
& \frac12\overline{a}_d^{pr} a_d^{sr}\left(6g_2^2\overline{y}_d^{st} y_d^{pt} + \overline{y}_u^{st} y_u^{pt}\left(2g_1^2+3g_2^2\right)\right) \mathrm{LF}_{2,1,1,0}[m_{\tilde q}^{s}, m_{\tilde d}^{r}, m_{\tilde q}^{p}] - \\
& \frac34\overline{a}_d^{pr} a_d^{sr}\left(2g_2^2\overline{y}_d^{st} y_d^{pt} + \overline{y}_u^{st} y_u^{pt}\left(g_1^2+g_2^2\right)\right) \mathrm{LF}_{3,1,1,-1}[m_{\tilde q}^{s}, m_{\tilde d}^{r}, m_{\tilde q}^{p}] + \\
& \frac12 g_1^2\tilde\mu^2\overline{y}_d^{pr} y_d^{sr}\overline{y}_u^{st} y_u^{pt}\, \mathrm{LF}_{2,1,1,0}[m_{\tilde q}^{s}, m_{\tilde d}^{r}, m_{\tilde u}^{t}] + \\
& \frac34\tilde\mu^2\overline{y}_d^{pr} y_d^{sr}\overline{y}_u^{st} y_u^{pt}\left(-g_1^2+g_2^2\right) \mathrm{LF}_{3,1,1,-1}[m_{\tilde q}^{s}, m_{\tilde d}^{r}, m_{\tilde u}^{t}] + \\
& \frac14\overline{y}_d^{st} y_d^{pt}\overline{a}_u^{pr} a_u^{sr}\left(g_1^2+3g_2^2\right) \mathrm{LF}_{2,1,1,0}[m_{\tilde q}^{s}, m_{\tilde q}^{p}, m_{\tilde u}^{r}] - \\
& \frac32 g_2^2\tilde\mu^2\overline{y}_d^{pr} y_d^{sr}\overline{y}_u^{st} y_u^{pt}\, \mathrm{LF}_{2,1,1,0}[m_{\tilde q}^{s}, m_{\tilde q}^{p}, m_{\tilde u}^{t}] + \\
& \frac32 g_2^2\tilde\mu^2\overline{y}_d^{pr} y_d^{sr}\overline{y}_u^{st} y_u^{pt}\, \mathrm{LF}_{3,1,1,-1}[m_{\tilde q}^{s}, m_{\tilde q}^{p}, m_{\tilde u}^{t}] + \\
& \frac34\tilde\mu^2\overline{y}_d^{pr} y_d^{sr}\overline{y}_u^{st} y_u^{pt}\left(-g_1^2+g_2^2\right) \mathrm{LF}_{2,2,1,-1}[m_{\tilde q}^{s}, m_{\tilde u}^{t}, m_{\tilde d}^{r}] + \\
& \frac34 g_2^2\tilde\mu^2\overline{y}_d^{pr} y_d^{sr}\overline{y}_u^{st} y_u^{pt}\, \mathrm{LF}_{2,2,1,-1}[m_{\tilde q}^{s}, m_{\tilde u}^{t}, m_{\tilde q}^{p}] + \\
& \frac12\overline{y}_d^{st} y_u^{sr} a_d^{pt}\overline{a}_u^{pr}\left(-2g_1^2+3g_2^2\right) \mathrm{LF}_{2,1,1,0}[m_{\tilde u}^{r}, m_{\tilde d}^{t}, m_{\tilde q}^{p}] - \\
& \frac32 g_2^2\overline{y}_d^{st} y_u^{sr} a_d^{pt}\overline{a}_u^{pr}\, \mathrm{LF}_{3,1,1,-1}[m_{\tilde u}^{r}, m_{\tilde d}^{t}, m_{\tilde q}^{p}] + \\
& \frac14\left(\overline{y}_d^{st} y_d^{pt}\overline{a}_u^{pr} a_u^{sr}\left(-7g_1^2+3g_2^2\right) + \tilde\mu^2\overline{y}_u^{pr}\overline{y}_u^{st} y_u^{pt} y_u^{sr}\left(g_1^2-3g_2^2\right)\right) \\
& \quad \mathrm{LF}_{2,1,1,0}[m_{\tilde u}^{r}, m_{\tilde q}^{p}, m_{\tilde q}^{s}] + \\
& \frac34\left(\overline{y}_d^{st} y_d^{pt}\overline{a}_u^{pr} a_u^{sr}\left(g_1^2-g_2^2\right) + \tilde\mu^2\overline{y}_u^{pr}\overline{y}_u^{st} y_u^{pt} y_u^{sr}\left(g_1^2+g_2^2\right)\right) \mathrm{LF}_{3,1,1,-1}[m_{\tilde u}^{r}, m_{\tilde q}^{p}, m_{\tilde q}^{s}] - \\
& \frac12 y_d^{sr}\overline{y}_u^{st}\left(2g_1^2-3g_2^2\right)\left(\overline{a}_d^{pr} a_u^{pt} - \tilde\mu^2\overline{y}_d^{pr} y_u^{pt}\right) \mathrm{LF}_{2,1,1,0}[m_{\tilde u}^{t}, m_{\tilde d}^{r}, m_{\tilde q}^{p}] + \\
& \frac32 g_2^2 y_d^{sr}\overline{y}_u^{st}\left(-\overline{a}_d^{pr} a_u^{pt} + \tilde\mu^2\overline{y}_d^{pr} y_u^{pt}\right) \mathrm{LF}_{3,1,1,-1}[m_{\tilde u}^{t}, m_{\tilde d}^{r}, m_{\tilde q}^{p}] + \\
& \frac12\tilde\mu^2\overline{y}_d^{pr} y_d^{sr}\overline{y}_u^{st} y_u^{pt}\left(2g_1^2-3g_2^2\right) \mathrm{LF}_{2,1,1,0}[m_{\tilde u}^{t}, m_{\tilde d}^{r}, m_{\tilde q}^{s}] + \\
& \frac32 g_2^2\tilde\mu^2\overline{y}_d^{pr} y_d^{sr}\overline{y}_u^{st} y_u^{pt}\, \mathrm{LF}_{3,1,1,-1}[m_{\tilde u}^{t}, m_{\tilde d}^{r}, m_{\tilde q}^{s}] + \frac34 g_1^2 g_2^4\, \mathrm{LF}_{2,1,1,-1}[\tilde\mu, m_1, m_2] - \\
& g_1^2 g_2^4\, \mathrm{LF}_{3,1,1,-2}[\tilde\mu, m_1, m_2] - 3g_1^2 g_2^4\tilde\mu^2\, \mathrm{LF}_{3,1,1,-1}[\tilde\mu, m_1, m_2] - \\
& g_1^4 g_2^2\, \mathrm{LF}_{3,2,1,-3}[\tilde\mu, m_1, m_2] - g_1^4 g_2^2\left(m_1^2+2m_1m_2+\tilde\mu^2\right) \mathrm{LF}_{3,2,1,-2}[\tilde\mu, m_1, m_2] + \\
& m_1 g_1^4 g_2^2\tilde\mu^2\left(m_1-2m_2\right) \mathrm{LF}_{3,2,1,-1}[\tilde\mu, m_1, m_2] + \frac12 g_1^2 g_2^4\, \mathrm{LF}_{4,1,1,-3}[\tilde\mu, m_1, m_2] + \\
& \frac52 g_1^2 g_2^4\tilde\mu^2\, \mathrm{LF}_{4,1,1,-2}[\tilde\mu, m_1, m_2] - g_1^2 g_2^4\left(m_2\left(m_1+m_2\right) - 3\tilde\mu^2\right) \mathrm{LF}_{3,2,1,-2}[\tilde\mu, m_2, m_1] - \\
& m_2 g_1^2 g_2^4\tilde\mu^2\left(2m_1+m_2\right) \mathrm{LF}_{3,2,1,-1}[\tilde\mu, m_2, m_1] - \\
& \frac12 g_1^2\tilde\mu^2\overline{y}_d^{st} y_d^{pt}\overline{a}_d^{pr} a_d^{sr}\, \mathrm{LF}_{2,1,1,1,0}[m_{\tilde d}^{r}, m_{\tilde d}^{t}, m_{\tilde q}^{p}, m_{\tilde q}^{s}] - \\
& 3\overline{y}_d^{st} y_d^{sr}\overline{y}_u^{uv} y_u^{pv}\overline{a}_d^{pr} a_d^{ut}\, \mathrm{LF}_{1,1,1,1,0}[m_{\tilde d}^{r}, m_{\tilde d}^{t}, m_{\tilde q}^{p}, m_{\tilde q}^{u}] - \\
& 3\overline{y}_d^{st}\overline{y}_u^{uv}\overline{a}_d^{pr}\left(y_d^{sr} y_d^{ut} a_u^{pv} + y_d^{ur} y_u^{sv} a_d^{pt}\right) \mathrm{LF}_{1,1,1,1,0}[m_{\tilde d}^{r}, m_{\tilde d}^{t}, m_{\tilde q}^{p}, m_{\tilde u}^{v}] + \\
& 3\tilde\mu^2\overline{y}_d^{pr}\overline{y}_d^{st}\overline{y}_d^{uv} y_d^{pt} y_d^{sv} y_d^{ur}\, \mathrm{LF}_{1,1,1,1,0}[m_{\tilde d}^{r}, m_{\tilde d}^{t}, m_{\tilde q}^{s}, m_{\tilde q}^{u}] + \\
& 3\tilde\mu^2\overline{y}_d^{pr}\overline{y}_d^{st} y_d^{pt} y_d^{ur}\overline{y}_u^{uv} y_u^{sv}\, \mathrm{LF}_{1,1,1,1,0}[m_{\tilde d}^{r}, m_{\tilde d}^{t}, m_{\tilde q}^{s}, m_{\tilde u}^{v}] + \\
& 3\tilde\mu^2\overline{y}_d^{pr}\overline{y}_d^{st} y_d^{pt} y_d^{ur}\overline{y}_u^{uv} y_u^{sv}\, \mathrm{LF}_{1,1,1,1,0}[m_{\tilde d}^{r}, m_{\tilde d}^{t}, m_{\tilde q}^{u}, m_{\tilde u}^{v}] - \\
& \frac38\tilde\mu^2\overline{y}_d^{st} y_d^{pt}\overline{a}_d^{pr} a_d^{sr}\left(g_1^2+g_2^2\right) \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde d}^{r}, m_{\tilde q}^{p}, m_{\tilde d}^{t}, m_{\tilde q}^{s}] - \\
& 3\overline{y}_d^{st}\left(\overline{y}_u^{uv}\overline{a}_d^{pr}\left(y_d^{ut} y_u^{pv} a_d^{sr} + y_d^{pt} y_u^{sv} a_d^{ur}\right) - \tilde\mu^2\overline{y}_d^{pr}\overline{y}_d^{uv} y_d^{pt} y_d^{sv} y_d^{ur}\right) \\
& \quad \mathrm{LF}_{1,1,1,1,0}[m_{\tilde d}^{r}, m_{\tilde q}^{p}, m_{\tilde q}^{s}, m_{\tilde q}^{u}] + \frac12 g_1^2 \\
& \quad \left(\tilde\mu^4\overline{y}_d^{pr} y_d^{sr}\overline{y}_u^{st} y_u^{pt} + \overline{a}_d^{pr} a_u^{pt}\left(a_d^{sr}\overline{a}_u^{st} - \tilde\mu^2 y_d^{sr}\overline{y}_u^{st}\right)\right) \mathrm{LF}_{2,1,1,1,0}[m_{\tilde d}^{r}, m_{\tilde q}^{p}, m_{\tilde q}^{s}, m_{\tilde u}^{t}] - \\
& 3\overline{y}_d^{st} y_d^{pt} y_d^{ur}\overline{y}_u^{uv}\overline{a}_d^{pr} a_u^{sv}\, \mathrm{LF}_{1,1,1,1,0}[m_{\tilde d}^{r}, m_{\tilde q}^{p}, m_{\tilde q}^{s}, m_{\tilde u}^{v}] - \\
& \frac38\tilde\mu^2\overline{y}_d^{st} y_d^{pt}\overline{a}_d^{pr} a_d^{sr}\left(g_1^2+g_2^2\right) \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde d}^{r}, m_{\tilde q}^{s}, m_{\tilde d}^{t}, m_{\tilde q}^{p}] - \\
& 3\tilde\mu^2\overline{y}_d^{pr} y_d^{sr}\overline{y}_u^{st}\overline{y}_u^{uv} y_u^{pv} y_u^{ut}\, \mathrm{LF}_{1,1,1,1,0}[m_{\tilde d}^{r}, m_{\tilde q}^{u}, m_{\tilde u}^{t}, m_{\tilde u}^{v}] - \\
& \frac12 g_1^2\tilde\mu^2\overline{y}_d^{st} y_d^{pt}\overline{a}_d^{pr} a_d^{sr}\, \mathrm{LF}_{2,1,1,1,0}[m_{\tilde d}^{t}, m_{\tilde d}^{r}, m_{\tilde q}^{p}, m_{\tilde q}^{s}] - \\
& 3\overline{y}_d^{st}\overline{y}_d^{uv} y_d^{sv} y_u^{ur} a_d^{pt}\overline{a}_u^{pr}\, \mathrm{LF}_{1,1,1,1,0}[m_{\tilde d}^{t}, m_{\tilde d}^{v}, m_{\tilde q}^{p}, m_{\tilde u}^{r}] + \\
& \frac38\tilde\mu^2\overline{y}_d^{st} y_d^{pt}\overline{a}_d^{pr} a_d^{sr}\left(g_1^2+g_2^2\right) \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde d}^{t}, m_{\tilde q}^{p}, m_{\tilde d}^{r}, m_{\tilde q}^{s}] - \\
& \frac12 g_1^2\tilde\mu^2\overline{y}_d^{st} y_u^{sr} a_d^{pt}\overline{a}_u^{pr}\, \mathrm{LF}_{2,1,1,1,0}[m_{\tilde d}^{t}, m_{\tilde q}^{p}, m_{\tilde q}^{s}, m_{\tilde u}^{r}] + \\
& \frac38\tilde\mu^2\overline{y}_d^{st} y_d^{pt}\overline{a}_d^{pr} a_d^{sr}\left(g_1^2+g_2^2\right) \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde d}^{t}, m_{\tilde q}^{s}, m_{\tilde d}^{r}, m_{\tilde q}^{p}] - \\
& 3\overline{y}_d^{st}\overline{y}_d^{uv} y_d^{pt} y_u^{ur} a_d^{sv}\overline{a}_u^{pr}\, \mathrm{LF}_{1,1,1,1,0}[m_{\tilde d}^{v}, m_{\tilde q}^{p}, m_{\tilde q}^{s}, m_{\tilde u}^{r}] - \\
& \frac12 g_1^2\tilde\mu^2\overline{y}_e^{st} y_e^{pt}\overline{a}_e^{pr} a_e^{sr}\, \mathrm{LF}_{2,1,1,1,0}[m_{\tilde e}^{r}, m_{\tilde e}^{t}, m_{\tilde l}^{p}, m_{\tilde l}^{s}] + \\
& \tilde\mu^2\overline{y}_e^{pr}\overline{y}_e^{st}\overline{y}_e^{uv} y_e^{pt} y_e^{sv} y_e^{ur}\, \mathrm{LF}_{1,1,1,1,0}[m_{\tilde e}^{r}, m_{\tilde e}^{t}, m_{\tilde l}^{s}, m_{\tilde l}^{u}] - \\
& \frac18\tilde\mu^2\overline{y}_e^{st} y_e^{pt}\overline{a}_e^{pr} a_e^{sr}\left(g_1^2+g_2^2\right) \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde e}^{r}, m_{\tilde l}^{p}, m_{\tilde e}^{t}, m_{\tilde l}^{s}] + \\
& \tilde\mu^2\overline{y}_e^{pr}\overline{y}_e^{st}\overline{y}_e^{uv} y_e^{pt} y_e^{sv} y_e^{ur}\, \mathrm{LF}_{1,1,1,1,0}[m_{\tilde e}^{r}, m_{\tilde l}^{p}, m_{\tilde l}^{s}, m_{\tilde l}^{u}] - \\
& \frac18\tilde\mu^2\overline{y}_e^{st} y_e^{pt}\overline{a}_e^{pr} a_e^{sr}\left(g_1^2+g_2^2\right) \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde e}^{r}, m_{\tilde l}^{s}, m_{\tilde e}^{t}, m_{\tilde l}^{p}] - \\
& \frac12 g_1^2\tilde\mu^2\overline{y}_e^{st} y_e^{pt}\overline{a}_e^{pr} a_e^{sr}\, \mathrm{LF}_{2,1,1,1,0}[m_{\tilde e}^{t}, m_{\tilde e}^{r}, m_{\tilde l}^{p}, m_{\tilde l}^{s}] + \\
& \frac18\tilde\mu^2\overline{y}_e^{st} y_e^{pt}\overline{a}_e^{pr} a_e^{sr}\left(g_1^2+g_2^2\right) \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde e}^{t}, m_{\tilde l}^{p}, m_{\tilde e}^{r}, m_{\tilde l}^{s}] + \\
& \frac18\tilde\mu^2\overline{y}_e^{st} y_e^{pt}\overline{a}_e^{pr} a_e^{sr}\left(g_1^2+g_2^2\right) \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde e}^{t}, m_{\tilde l}^{s}, m_{\tilde e}^{r}, m_{\tilde l}^{p}] + \\
& \frac14\overline{a}_e^{pr} a_e^{sr}\left(4g_2^2\overline{a}_e^{st} a_e^{pt} + \tilde\mu^2\overline{y}_e^{st} y_e^{pt}\left(g_1^2-g_2^2\right)\right) \mathrm{LF}_{2,1,1,1,0}[m_{\tilde l}^{p}, m_{\tilde e}^{r}, m_{\tilde e}^{t}, m_{\tilde l}^{s}] - \\
& \frac12 g_2^2\overline{a}_e^{pr}\overline{a}_e^{st} a_e^{pt} a_e^{sr}\, \mathrm{LF}_{3,1,1,1,-1}[m_{\tilde l}^{p}, m_{\tilde e}^{r}, m_{\tilde e}^{t}, m_{\tilde l}^{s}] + \\
& \frac14\tilde\mu^2\overline{y}_e^{st} y_e^{pt}\overline{a}_e^{pr} a_e^{sr}\left(g_1^2-g_2^2\right) \mathrm{LF}_{2,1,1,1,0}[m_{\tilde l}^{s}, m_{\tilde e}^{r}, m_{\tilde e}^{t}, m_{\tilde l}^{p}] - \\
& \frac12 g_2^2\tilde\mu^2\overline{y}_e^{st} y_e^{sr}\overline{a}_e^{pr} a_e^{pt}\, \mathrm{LF}_{3,1,1,1,-1}[m_{\tilde l}^{s}, m_{\tilde e}^{r}, m_{\tilde e}^{t}, m_{\tilde l}^{p}] + \\
& \frac14\overline{a}_d^{pr} a_d^{sr}\left(12g_2^2\overline{a}_d^{st} a_d^{pt} - \tilde\mu^2\overline{y}_d^{st} y_d^{pt}\left(g_1^2+3g_2^2\right)\right) \mathrm{LF}_{2,1,1,1,0}[m_{\tilde q}^{p}, m_{\tilde d}^{r}, m_{\tilde d}^{t}, m_{\tilde q}^{s}] - \\
& \frac32 g_2^2\overline{a}_d^{pr}\overline{a}_d^{st} a_d^{pt} a_d^{sr}\, \mathrm{LF}_{3,1,1,1,-1}[m_{\tilde q}^{p}, m_{\tilde d}^{r}, m_{\tilde d}^{t}, m_{\tilde q}^{s}] + \\
& \frac12\left(-g_1^2\tilde\mu^4\overline{y}_d^{pr} y_d^{sr}\overline{y}_u^{st} y_u^{pt} + \overline{a}_d^{pr} a_u^{pt}\left(2g_1^2+3g_2^2\right)\left(a_d^{sr}\overline{a}_u^{st} - \tilde\mu^2 y_d^{sr}\overline{y}_u^{st}\right)\right) \\
& \quad \mathrm{LF}_{2,1,1,1,0}[m_{\tilde q}^{p}, m_{\tilde d}^{r}, m_{\tilde q}^{s}, m_{\tilde u}^{t}] - \\
& \frac34\left(\tilde\mu^4\overline{y}_d^{pr} y_d^{sr}\overline{y}_u^{st} y_u^{pt}\left(-g_1^2+g_2^2\right) + \overline{a}_d^{pr} a_u^{pt}\left(g_1^2+g_2^2\right)\left(a_d^{sr}\overline{a}_u^{st} - \tilde\mu^2 y_d^{sr}\overline{y}_u^{st}\right)\right) \\
& \quad \mathrm{LF}_{3,1,1,1,-1}[m_{\tilde q}^{p}, m_{\tilde d}^{r}, m_{\tilde q}^{s}, m_{\tilde u}^{t}] - \\
& \frac12\tilde\mu^2\overline{y}_d^{st} y_u^{sr} a_d^{pt}\overline{a}_u^{pr}\left(2g_1^2+3g_2^2\right) \mathrm{LF}_{2,1,1,1,0}[m_{\tilde q}^{p}, m_{\tilde d}^{t}, m_{\tilde q}^{s}, m_{\tilde u}^{r}] + \\
& \frac34\tilde\mu^2\overline{y}_d^{st} y_u^{sr} a_d^{pt}\overline{a}_u^{pr}\left(g_1^2+g_2^2\right) \mathrm{LF}_{3,1,1,1,-1}[m_{\tilde q}^{p}, m_{\tilde d}^{t}, m_{\tilde q}^{s}, m_{\tilde u}^{r}] - \\
& \frac34\left(\tilde\mu^4\overline{y}_d^{pr} y_d^{sr}\overline{y}_u^{st} y_u^{pt}\left(-g_1^2+g_2^2\right) + \overline{a}_d^{pr} a_d^{sr}\left(\overline{a}_u^{st} a_u^{pt}\left(g_1^2+g_2^2\right) + 2g_2^2\tilde\mu^2\overline{y}_u^{st} y_u^{pt}\right)\right) \\
& \quad \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde q}^{p}, m_{\tilde q}^{s}, m_{\tilde d}^{r}, m_{\tilde u}^{t}] - \\
& 3\overline{y}_d^{st}\overline{y}_d^{uv} y_d^{pv} y_d^{ut}\overline{a}_u^{pr} a_u^{sr}\, \mathrm{LF}_{1,1,1,1,0}[m_{\tilde q}^{p}, m_{\tilde q}^{s}, m_{\tilde q}^{u}, m_{\tilde u}^{r}] + \\
& \frac14\tilde\mu^2\overline{y}_u^{st}\overline{a}_u^{pr}\left(-y_u^{pt} a_u^{sr}\left(g_1^2-3g_2^2\right) + 9g_2^2 y_u^{sr} a_u^{pt}\right) \mathrm{LF}_{2,1,1,1,0}[m_{\tilde q}^{p}, m_{\tilde q}^{s}, m_{\tilde u}^{r}, m_{\tilde u}^{t}] - \\
& \frac34 g_2^2\tilde\mu^2\overline{y}_u^{st} y_u^{sr}\left(\overline{a}_u^{pr} a_u^{pt} + 2\tilde\mu^2\overline{y}_u^{pr} y_u^{pt}\right) \mathrm{LF}_{3,1,1,1,-1}[m_{\tilde q}^{p}, m_{\tilde q}^{s}, m_{\tilde u}^{r}, m_{\tilde u}^{t}] + \\
& \frac34\tilde\mu^2\overline{y}_d^{st} y_u^{sr} a_d^{pt}\overline{a}_u^{pr}\left(g_1^2+g_2^2\right) \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde q}^{p}, m_{\tilde u}^{r}, m_{\tilde d}^{t}, m_{\tilde q}^{s}] - \\
& \frac34 g_2^2\tilde\mu^4\overline{y}_u^{pr}\overline{y}_u^{st} y_u^{pt} y_u^{sr}\, \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde q}^{p}, m_{\tilde u}^{r}, m_{\tilde q}^{s}, m_{\tilde u}^{t}] - \\
& \frac34\left(\tilde\mu^4\overline{y}_d^{pr} y_d^{sr}\overline{y}_u^{st} y_u^{pt}\left(-g_1^2+g_2^2\right) + \overline{a}_d^{pr} a_u^{pt}\left(g_1^2+g_2^2\right)\left(a_d^{sr}\overline{a}_u^{st} - \tilde\mu^2 y_d^{sr}\overline{y}_u^{st}\right)\right) \\
& \quad \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde q}^{p}, m_{\tilde u}^{t}, m_{\tilde d}^{r}, m_{\tilde q}^{s}] - \frac34 g_2^2\tilde\mu^4\overline{y}_u^{pr}\overline{y}_u^{st} y_u^{pt} y_u^{sr}\, \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde q}^{p}, m_{\tilde u}^{t}, m_{\tilde q}^{s}, m_{\tilde u}^{r}] - \\
& \frac14\tilde\mu^2\overline{y}_d^{st} y_d^{pt}\overline{a}_d^{pr} a_d^{sr}\left(g_1^2+3g_2^2\right) \mathrm{LF}_{2,1,1,1,0}[m_{\tilde q}^{s}, m_{\tilde d}^{r}, m_{\tilde d}^{t}, m_{\tilde q}^{p}] - \\
& \frac32 g_2^2\tilde\mu^2\overline{y}_d^{st} y_d^{sr}\overline{a}_d^{pr} a_d^{pt}\, \mathrm{LF}_{3,1,1,1,-1}[m_{\tilde q}^{s}, m_{\tilde d}^{r}, m_{\tilde d}^{t}, m_{\tilde q}^{p}] + \\
& \frac12\left(-g_1^2\tilde\mu^4\overline{y}_d^{pr} y_d^{sr}\overline{y}_u^{st} y_u^{pt} + \overline{a}_d^{pr} a_u^{pt}\left(a_d^{sr}\overline{a}_u^{st}\left(2g_1^2+3g_2^2\right) + g_1^2\tilde\mu^2 y_d^{sr}\overline{y}_u^{st}\right)\right) \\
& \quad \mathrm{LF}_{2,1,1,1,0}[m_{\tilde q}^{s}, m_{\tilde d}^{r}, m_{\tilde q}^{p}, m_{\tilde u}^{t}] - \\
& \frac34\left(\tilde\mu^4\overline{y}_d^{pr} y_d^{sr}\overline{y}_u^{st} y_u^{pt}\left(-g_1^2+g_2^2\right) + \overline{a}_d^{pr} a_u^{pt}\left(a_d^{sr}\overline{a}_u^{st}\left(g_1^2+g_2^2\right) + \tilde\mu^2 y_d^{sr}\overline{y}_u^{st}\left(g_1^2-g_2^2\right)\right)\right) \\
& \quad \mathrm{LF}_{3,1,1,1,-1}[m_{\tilde q}^{s}, m_{\tilde d}^{r}, m_{\tilde q}^{p}, m_{\tilde u}^{t}] + \frac12 g_1^2\tilde\mu^2\overline{y}_d^{st} y_u^{sr} a_d^{pt}\overline{a}_u^{pr}\, \mathrm{LF}_{2,1,1,1,0}[m_{\tilde q}^{s}, m_{\tilde d}^{t}, m_{\tilde q}^{p}, m_{\tilde u}^{r}] + \\
& \frac34\tilde\mu^2\overline{y}_d^{st} y_u^{sr} a_d^{pt}\overline{a}_u^{pr}\left(-g_1^2+g_2^2\right) \mathrm{LF}_{3,1,1,1,-1}[m_{\tilde q}^{s}, m_{\tilde d}^{t}, m_{\tilde q}^{p}, m_{\tilde u}^{r}] + \\
& \frac14\tilde\mu^2\overline{y}_u^{st}\overline{a}_u^{pr}\left(-y_u^{pt} a_u^{sr}\left(g_1^2-3g_2^2\right) + 3g_2^2 y_u^{sr} a_u^{pt}\right) \mathrm{LF}_{2,1,1,1,0}[m_{\tilde q}^{s}, m_{\tilde q}^{p}, m_{\tilde u}^{r}, m_{\tilde u}^{t}] - \\
& \frac34 g_2^2\tilde\mu^2\overline{y}_u^{st} y_u^{sr}\overline{a}_u^{pr} a_u^{pt}\, \mathrm{LF}_{3,1,1,1,-1}[m_{\tilde q}^{s}, m_{\tilde q}^{p}, m_{\tilde u}^{r}, m_{\tilde u}^{t}] + \\
& \frac34\tilde\mu^2\overline{y}_d^{st} y_u^{sr} a_d^{pt}\overline{a}_u^{pr}\left(-g_1^2+g_2^2\right) \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde q}^{s}, m_{\tilde u}^{r}, m_{\tilde d}^{t}, m_{\tilde q}^{p}] - \\
& \frac34 g_2^2\tilde\mu^2\overline{y}_u^{st} y_u^{sr}\overline{a}_u^{pr} a_u^{pt}\, \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde q}^{s}, m_{\tilde u}^{r}, m_{\tilde q}^{p}, m_{\tilde u}^{t}] - \\
& \frac34\left(\tilde\mu^4\overline{y}_d^{pr} y_d^{sr}\overline{y}_u^{st} y_u^{pt}\left(-g_1^2+g_2^2\right) + \overline{a}_d^{pr} a_u^{pt}\left(a_d^{sr}\overline{a}_u^{st}\left(g_1^2+g_2^2\right) + \tilde\mu^2 y_d^{sr}\overline{y}_u^{st}\left(g_1^2-g_2^2\right)\right)\right) \\
& \quad \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde q}^{s}, m_{\tilde u}^{t}, m_{\tilde d}^{r}, m_{\tilde q}^{p}] - \frac34 g_2^2\tilde\mu^2\overline{y}_u^{st} y_u^{sr}\overline{a}_u^{pr} a_u^{pt}\, \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde q}^{s}, m_{\tilde u}^{t}, m_{\tilde q}^{p}, m_{\tilde u}^{r}] + \\
& \frac12\tilde\mu^2\overline{y}_d^{st} y_u^{sr} a_d^{pt}\overline{a}_u^{pr}\left(2g_1^2-3g_2^2\right) \mathrm{LF}_{2,1,1,1,0}[m_{\tilde u}^{r}, m_{\tilde d}^{t}, m_{\tilde q}^{p}, m_{\tilde q}^{s}] + \\
& \frac32 g_2^2\tilde\mu^2\overline{y}_d^{st} y_u^{sr} a_d^{pt}\overline{a}_u^{pr}\, \mathrm{LF}_{3,1,1,1,-1}[m_{\tilde u}^{r}, m_{\tilde d}^{t}, m_{\tilde q}^{p}, m_{\tilde q}^{s}] + \\
& \frac14\tilde\mu^2\overline{y}_u^{st}\left(\overline{a}_u^{pr}\left(y_u^{pt} a_u^{sr}\left(7g_1^2+3g_2^2\right) + 3g_2^2 y_u^{sr} a_u^{pt}\right) + 6g_2^2\tilde\mu^2\overline{y}_u^{pr} y_u^{pt} y_u^{sr}\right) \\
& \quad \mathrm{LF}_{2,1,1,1,0}[m_{\tilde u}^{r}, m_{\tilde q}^{p}, m_{\tilde q}^{s}, m_{\tilde u}^{t}] - \\
& \frac34\tilde\mu^2\overline{y}_u^{st}\left(\overline{a}_u^{pr}\left(y_u^{pt} a_u^{sr}\left(g_1^2+g_2^2\right) + g_2^2 y_u^{sr} a_u^{pt}\right) + 2g_2^2\tilde\mu^2\overline{y}_u^{pr} y_u^{pt} y_u^{sr}\right) \\
& \quad \mathrm{LF}_{3,1,1,1,-1}[m_{\tilde u}^{r}, m_{\tilde q}^{p}, m_{\tilde q}^{s}, m_{\tilde u}^{t}] - \\
& \frac34 g_2^2\tilde\mu^2\overline{y}_u^{st} y_u^{sr}\left(\overline{a}_u^{pr} a_u^{pt} + \tilde\mu^2\overline{y}_u^{pr} y_u^{pt}\right) \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde u}^{r}, m_{\tilde u}^{t}, m_{\tilde q}^{p}, m_{\tilde q}^{s}] - \\
& \frac12\left(2g_1^2-3g_2^2\right)\left(\tilde\mu^4\overline{y}_d^{pr} y_d^{sr}\overline{y}_u^{st} y_u^{pt} + \overline{a}_d^{pr} a_u^{pt}\left(a_d^{sr}\overline{a}_u^{st} - \tilde\mu^2 y_d^{sr}\overline{y}_u^{st}\right)\right) \\
& \quad \mathrm{LF}_{2,1,1,1,0}[m_{\tilde u}^{t}, m_{\tilde d}^{r}, m_{\tilde q}^{p}, m_{\tilde q}^{s}] - \frac32 g_2^2 \\
& \quad \left(\tilde\mu^4\overline{y}_d^{pr} y_d^{sr}\overline{y}_u^{st} y_u^{pt} + \overline{a}_d^{pr} a_u^{pt}\left(a_d^{sr}\overline{a}_u^{st} - \tilde\mu^2 y_d^{sr}\overline{y}_u^{st}\right)\right) \mathrm{LF}_{3,1,1,1,-1}[m_{\tilde u}^{t}, m_{\tilde d}^{r}, m_{\tilde q}^{p}, m_{\tilde q}^{s}] + \\
& \frac14\tilde\mu^2\overline{y}_u^{st}\overline{a}_u^{pr}\left(y_u^{pt} a_u^{sr}\left(g_1^2-3g_2^2\right) + 3g_2^2 y_u^{sr} a_u^{pt}\right) \mathrm{LF}_{2,1,1,1,0}[m_{\tilde u}^{t}, m_{\tilde q}^{p}, m_{\tilde q}^{s}, m_{\tilde u}^{r}] + \\
& \frac34\tilde\mu^2\overline{y}_u^{st}\overline{a}_u^{pr}\left(y_u^{pt} a_u^{sr}\left(g_1^2+g_2^2\right) - g_2^2 y_u^{sr} a_u^{pt}\right) \mathrm{LF}_{3,1,1,1,-1}[m_{\tilde u}^{t}, m_{\tilde q}^{p}, m_{\tilde q}^{s}, m_{\tilde u}^{r}] + \\
& 3\tilde\mu^2\overline{y}_d^{st}\overline{y}_d^{uv}\overline{a}_d^{pr}\left(y_d^{pt} y_d^{ur} a_d^{sv} + y_d^{pv} y_d^{ut} a_d^{sr}\right) \mathrm{LF}_{1,1,1,1,1,0}[m_{\tilde d}^{r}, m_{\tilde d}^{t}, m_{\tilde d}^{v}, m_{\tilde q}^{p}, m_{\tilde q}^{s}] + \\
& 3\overline{a}_d^{pr}\left(-\overline{y}_u^{uv} y_u^{sv}\overline{a}_d^{st} a_d^{pt} a_d^{ur} + \tilde\mu^2\overline{y}_d^{st}\overline{y}_d^{uv}\left(y_d^{sv} y_d^{ur} a_d^{pt} + y_d^{pv} y_d^{ut} a_d^{sr}\right)\right) \\
& \quad \mathrm{LF}_{1,1,1,1,1,0}[m_{\tilde d}^{r}, m_{\tilde d}^{t}, m_{\tilde q}^{p}, m_{\tilde q}^{s}, m_{\tilde q}^{u}] + \\
& 3 y_d^{ur}\overline{y}_u^{uv}\overline{a}_d^{pr} a_d^{pt}\left(-\overline{a}_d^{st} a_u^{sv} + \tilde\mu^2\overline{y}_d^{st} y_u^{sv}\right) \mathrm{LF}_{1,1,1,1,1,0}[m_{\tilde d}^{r}, m_{\tilde d}^{t}, m_{\tilde q}^{p}, m_{\tilde q}^{s}, m_{\tilde u}^{v}] + \\
& 3\tilde\mu^2\overline{y}_d^{st}\overline{y}_u^{uv}\overline{a}_d^{pr}\left(y_d^{sr} y_d^{ut} a_u^{pv} + y_d^{ur} y_u^{sv} a_d^{pt}\right) \mathrm{LF}_{1,1,1,1,1,0}[m_{\tilde d}^{r}, m_{\tilde d}^{t}, m_{\tilde q}^{p}, m_{\tilde q}^{u}, m_{\tilde u}^{v}] - \\
& 3\tilde\mu^4\overline{y}_d^{pr}\overline{y}_d^{st} y_d^{pt} y_d^{ur}\overline{y}_u^{uv} y_u^{sv}\, \mathrm{LF}_{1,1,1,1,1,0}[m_{\tilde d}^{r}, m_{\tilde d}^{t}, m_{\tilde q}^{s}, m_{\tilde q}^{u}, m_{\tilde u}^{v}] + \\
& 3\tilde\mu^2\overline{y}_d^{st}\overline{y}_d^{uv} y_d^{pv} y_d^{ut}\overline{a}_d^{pr} a_d^{sr}\, \mathrm{LF}_{1,1,1,1,1,0}[m_{\tilde d}^{r}, m_{\tilde d}^{v}, m_{\tilde q}^{p}, m_{\tilde q}^{s}, m_{\tilde q}^{u}] - \\
& 3\overline{y}_d^{uv}\overline{a}_d^{pr}\overline{a}_u^{st}\left(y_d^{ur} a_d^{sv} a_u^{pt} + y_u^{ut} a_d^{pv} a_d^{sr}\right) \mathrm{LF}_{1,1,1,1,1,0}[m_{\tilde d}^{r}, m_{\tilde d}^{v}, m_{\tilde q}^{p}, m_{\tilde q}^{s}, m_{\tilde u}^{t}] - \\
& 3\overline{y}_d^{uv}\overline{a}_d^{pr}\overline{a}_u^{st}\left(y_d^{pv} a_d^{sr} a_u^{ut} + y_d^{sv} a_d^{ur} a_u^{pt}\right) \mathrm{LF}_{1,1,1,1,1,0}[m_{\tilde d}^{r}, m_{\tilde q}^{p}, m_{\tilde q}^{s}, m_{\tilde q}^{u}, m_{\tilde u}^{t}] + \\
& 3\tilde\mu^2\overline{y}_d^{st} y_d^{pt} y_d^{ur}\overline{y}_u^{uv}\overline{a}_d^{pr} a_u^{sv}\, \mathrm{LF}_{1,1,1,1,1,0}[m_{\tilde d}^{r}, m_{\tilde q}^{p}, m_{\tilde q}^{s}, m_{\tilde q}^{u}, m_{\tilde u}^{v}] + \\
& 3\tilde\mu^2 y_d^{sr}\overline{y}_u^{st}\overline{y}_u^{uv} y_u^{ut}\left(-\overline{a}_d^{pr} a_u^{pv} + \tilde\mu^2\overline{y}_d^{pr} y_u^{pv}\right) \mathrm{LF}_{1,1,1,1,1,0}[m_{\tilde d}^{r}, m_{\tilde q}^{p}, m_{\tilde q}^{u}, m_{\tilde u}^{t}, m_{\tilde u}^{v}] + \\
& 3\tilde\mu^4\overline{y}_d^{pr} y_d^{sr}\overline{y}_u^{st}\overline{y}_u^{uv} y_u^{pv} y_u^{ut}\, \mathrm{LF}_{1,1,1,1,1,0}[m_{\tilde d}^{r}, m_{\tilde q}^{s}, m_{\tilde q}^{u}, m_{\tilde u}^{t}, m_{\tilde u}^{v}] + \\
& 3\tilde\mu^2\overline{y}_d^{st}\overline{y}_d^{uv} y_d^{sv} y_u^{ur} a_d^{pt}\overline{a}_u^{pr}\, \mathrm{LF}_{1,1,1,1,1,0}[m_{\tilde d}^{t}, m_{\tilde d}^{v}, m_{\tilde q}^{p}, m_{\tilde q}^{u}, m_{\tilde u}^{r}] - \\
& 3\tilde\mu^2\overline{y}_d^{st}\overline{y}_u^{uv} y_u^{sv} y_u^{ur} a_d^{pt}\overline{a}_u^{pr}\, \mathrm{LF}_{1,1,1,1,1,0}[m_{\tilde d}^{t}, m_{\tilde q}^{p}, m_{\tilde q}^{u}, m_{\tilde u}^{r}, m_{\tilde u}^{v}] + \\
& 3\tilde\mu^2\overline{y}_d^{st}\overline{y}_d^{uv} y_d^{pt} y_u^{ur} a_d^{sv}\overline{a}_u^{pr}\, \mathrm{LF}_{1,1,1,1,1,0}[m_{\tilde d}^{v}, m_{\tilde q}^{p}, m_{\tilde q}^{s}, m_{\tilde q}^{u}, m_{\tilde u}^{r}] + \\
& \tilde\mu^2\overline{y}_e^{st}\overline{y}_e^{uv}\overline{a}_e^{pr}\left(y_e^{pt} y_e^{ur} a_e^{sv} + y_e^{pv} y_e^{ut} a_e^{sr}\right) \mathrm{LF}_{1,1,1,1,1,0}[m_{\tilde e}^{r}, m_{\tilde e}^{t}, m_{\tilde e}^{v}, m_{\tilde l}^{p}, m_{\tilde l}^{s}] + \\
& \tilde\mu^2\overline{y}_e^{st}\overline{y}_e^{uv}\overline{a}_e^{pr}\left(y_e^{sv} y_e^{ur} a_e^{pt} + y_e^{pv} y_e^{ut} a_e^{sr}\right) \mathrm{LF}_{1,1,1,1,1,0}[m_{\tilde e}^{r}, m_{\tilde e}^{t}, m_{\tilde l}^{p}, m_{\tilde l}^{s}, m_{\tilde l}^{u}] + \\
& \tilde\mu^2\overline{y}_e^{st}\overline{y}_e^{uv} y_e^{pv} y_e^{ut}\overline{a}_e^{pr} a_e^{sr}\, \mathrm{LF}_{1,1,1,1,1,0}[m_{\tilde e}^{r}, m_{\tilde e}^{v}, m_{\tilde l}^{p}, m_{\tilde l}^{s}, m_{\tilde l}^{u}] - \\
& 3\tilde\mu^2\overline{y}_d^{st} y_d^{pt}\overline{y}_u^{uv} y_u^{ur}\overline{a}_u^{pr} a_u^{sv}\, \mathrm{LF}_{1,1,1,1,1,0}[m_{\tilde q}^{p}, m_{\tilde q}^{s}, m_{\tilde q}^{u}, m_{\tilde u}^{r}, m_{\tilde u}^{v}] + \\
& 3\tilde\mu^4\overline{y}_u^{pr}\overline{y}_u^{st}\overline{y}_u^{uv} y_u^{pt} y_u^{sv} y_u^{ur}\, \mathrm{LF}_{1,1,1,1,1,0}[m_{\tilde q}^{p}, m_{\tilde q}^{s}, m_{\tilde q}^{u}, m_{\tilde u}^{t}, m_{\tilde u}^{v}] + \\
& 3\tilde\mu^2\overline{y}_d^{uv}\overline{a}_d^{pr}\overline{a}_d^{st}\left(y_d^{sr} a_d^{pv} a_d^{ut} + y_d^{sv} a_d^{pt} a_d^{ur}\right) \mathrm{LF}_{1,1,1,1,1,1,0}[m_{\tilde d}^{r}, m_{\tilde d}^{t}, m_{\tilde d}^{v}, m_{\tilde q}^{p}, m_{\tilde q}^{s}, m_{\tilde q}^{u}] - \\
& 3\overline{a}_d^{pr} a_d^{pt}\left(\tilde\mu^4\overline{y}_d^{st} y_d^{ur}\overline{y}_u^{uv} y_u^{sv} + \overline{a}_d^{st} a_u^{sv}\left(a_d^{ur}\overline{a}_u^{uv} - \tilde\mu^2 y_d^{ur}\overline{y}_u^{uv}\right)\right) \\
& \quad \mathrm{LF}_{1,1,1,1,1,1,0}[m_{\tilde d}^{r}, m_{\tilde d}^{t}, m_{\tilde q}^{p}, m_{\tilde q}^{s}, m_{\tilde q}^{u}, m_{\tilde u}^{v}] + \\
& 3\tilde\mu^2\overline{y}_d^{uv} y_u^{ut}\overline{a}_d^{pr} a_d^{pv} a_d^{sr}\overline{a}_u^{st}\, \mathrm{LF}_{1,1,1,1,1,1,0}[m_{\tilde d}^{r}, m_{\tilde d}^{v}, m_{\tilde q}^{p}, m_{\tilde q}^{s}, m_{\tilde q}^{u}, m_{\tilde u}^{t}] - \\
& 3\tilde\mu^2\overline{y}_u^{uv}\left(\tilde\mu^4\overline{y}_d^{pr} y_d^{sr}\overline{y}_u^{st} y_u^{pv} y_u^{ut} + \overline{a}_d^{pr}\left(y_u^{ut} a_d^{sr}\overline{a}_u^{st} a_u^{pv} - \tilde\mu^2 y_d^{ur}\overline{y}_u^{st} y_u^{sv} a_u^{pt}\right)\right) \\
& \quad \mathrm{LF}_{1,1,1,1,1,1,0}[m_{\tilde d}^{r}, m_{\tilde q}^{p}, m_{\tilde q}^{s}, m_{\tilde q}^{u}, m_{\tilde u}^{t}, m_{\tilde u}^{v}] + \\
& 3\tilde\mu^4\overline{y}_d^{st}\overline{y}_u^{uv} y_u^{sv} y_u^{ur} a_d^{pt}\overline{a}_u^{pr}\, \mathrm{LF}_{1,1,1,1,1,1,0}[m_{\tilde d}^{t}, m_{\tilde q}^{p}, m_{\tilde q}^{s}, m_{\tilde q}^{u}, m_{\tilde u}^{r}, m_{\tilde u}^{v}] + \\
& \tilde\mu^2\overline{y}_e^{uv}\overline{a}_e^{pr}\overline{a}_e^{st}\left(y_e^{sr} a_e^{pv} a_e^{ut} + y_e^{sv} a_e^{pt} a_e^{ur}\right) \mathrm{LF}_{1,1,1,1,1,1,0}[m_{\tilde e}^{r}, m_{\tilde e}^{t}, m_{\tilde e}^{v}, m_{\tilde l}^{p}, m_{\tilde l}^{s}, m_{\tilde l}^{u}] + \\
& 3\tilde\mu^4\overline{y}_u^{st}\overline{y}_u^{uv}\overline{a}_u^{pr}\left(y_u^{pt} y_u^{ur} a_u^{sv} + y_u^{pv} y_u^{ut} a_u^{sr}\right) \mathrm{LF}_{1,1,1,1,1,1,0}[m_{\tilde q}^{p}, m_{\tilde q}^{s}, m_{\tilde q}^{u}, m_{\tilde u}^{r}, m_{\tilde u}^{t}, m_{\tilde u}^{v}]\Big)
\end{aligned}
$$